JOHN ZHANG

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EDUCATION

UNIVERSITY OF TORONTO Toronto, ON
Bachelor of Science Expected Jun 2028
Major in Computer Science; Major in Statistical Science
Cumulative GPA: 3.89/4.0; Innis College Anniversary Scholarship 2025, Dean's List Scholar 2025
Relevant Coursework: Machine Learning, Data Analysis, Statistics, Deep Learning, Data Structures & Algorithms

WORK EXPERIENCE

SUNNYBROOK RESEARCH INSTITUTE Toronto, ON
Machine Learning Analyst May 2025 – Present

- Developed CNN and LSTM models to map modern accelerometer measurements to legacy Actical counts, enabling reuse of historical data in clinical research.
- Developed and benchmarked sleep–wake classification models using large-scale actigraphy data; designed and containerized a scalable processing pipeline with Docker for reproducible analysis of UK Biobank data.
- Developed a SuperAger classification model using activity-derived features to investigate patterns associated with healthy aging.
- Conducted exploratory data analysis, feature engineering, and signal processing across EEG and actigraphy datasets, including spectral analysis and time-frequency feature extraction.
- Performed quantitative model evaluation and error analysis using subject-level validation and performance metrics to assess generalization and guide model improvements.

UNIVERSITY OF TORONTO AEROSPACE TEAM Toronto, ON
Data Processing Engineer Sep 2024 – Jan 2025

- Debugged and modernized legacy Python and SQL code while researching hyperspectral image correction methods, restoring satellite-data processing capabilities and improving image accuracy for scientific analysis.

PROJECTS

COST-AWARE TOOL-USE LEARNING FOR LLM GitHub Repo | Sep 2026 – Present

- Built a Python-based tool-use environment with 35 priced tools across 10 overlapping cost families, using Pydantic-typed state and structured JSON outputs to support multi-step agent decisions.
- Generated 7.7K synthetic tasks from 50 templates, producing 11K supervised fine-tuning demonstrations and 2.8K step-level DPO preference pairs through automated execution of valid and invalid tool paths.
- Fine-tuned Qwen2.5-1.5B using LoRA SFT and DPO on an H100 via Slurm, improving held-out task success to 98% and reducing excess tool spending from 29% to 11%.

AUTOML AGENT Live Demo | GitHub Repo | Jul 2026 – Present

- Architected a multi-agent LLM pipeline using LangGraph that converts natural-language ML requests into trained models by automating dataset analysis, experiment planning, model generation, and result reporting.
- Developed a retrieval-augmented research agent using Qdrant to ground experimentation decisions in relevant documentation, alongside a self-repairing agent that diagnoses and resolves Python training-code failures.
- Deployed on GCP using Cloud Run, Firestore, and Cloud Storage, integrating live prediction serving, drift monitoring (Evidently), automated retraining, and LangSmith evaluation in CI to detect agent regressions.

TECHNICAL SKILLS

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face (Transformers, PEFT, TRL), LoRA/QLoRA Fine-Tuning, DPO, LLM Evaluation (LLM-as-Judge, LangSmith), Evidently, TensorBoard
Engineering: Python, Pydantic, FastAPI, LangGraph, RAG, Qdrant, Docker, Git, CI/CD (GitHub Actions)
Data/Cloud: NumPy, Pandas, SciPy, SQL, Google Cloud Platform, Signal Processing